

ASME and Environmental Responsibilities

1. Introduction

- ASME gives engineering standards that ensure **safety, reliability, and quality**.
- These standards also support **sustainability and protection of natural systems**.
- In short, ASME is not only about design rules, but also about **ethical responsibility**.

2. ASME in simple words

- ASME stands for **American Society of Mechanical Engineers**.
- It provides **codes and standards** for boilers, pressure vessels, piping, elevators, and other mechanical systems.
- It helps engineers build safe and efficient systems.

3. Safety and environment link

- Safe design prevents **leaks, explosions, and failures**.
- If a pressure vessel or boiler fails, it can cause **pollution, fire, toxic release, and land/water contamination**.
- So, **safety = protection of people + protection of environment**.

4. Sustainability link

- ASME standards help equipment last longer.
- Long life means **less waste, less material use, and lower carbon footprint**.
- Efficient boilers, turbines, and heat systems reduce **fuel consumption** and **CO2 emissions**.

5. Protection of natural systems

- Leak testing and proper design reduce **air, water, and soil pollution**.
- Strong piping and vessel standards prevent **fugitive emissions** and chemical leakage.
- This helps protect ecosystems, wildlife, and groundwater.

6. Energy project example

- In **supercritical and ultra-supercritical power plants**, ASME boiler standards are very important.
- These plants work at high pressure and temperature for better efficiency.
- Higher efficiency means **less fuel burned** and **less greenhouse gas emission**.

7. Manufacturing project example

- In **chemical pressure vessels**, ASME BPVC Section VIII is used.
- **Design by Analysis** can reduce unnecessary material while keeping safety.
- This saves steel, lowers cost, reduces transport energy, and cuts embodied carbon.

8. Main environmental responsibilities of engineers under ASME

- Design for **safety and containment**.
- Reduce **emissions and leakage**.
- Improve **energy efficiency**.

- Use **less material without reducing safety**.
- Follow **codes, inspections, and testing** properly.

9. Ethical importance

- Engineers must not compromise safety for profit or speed.
- They should protect **human life, environment, and natural resources**.
- Ethical engineering means **safe + efficient + sustainable design**.

10. Conclusion

- ASME standards support **safe engineering, sustainable development, and environmental protection**.
- They help engineers make systems that are **reliable, efficient, and eco-friendly**.
- Therefore, ASME is a key guide for responsible engineering practice.

Very short memory line

ASME = Safety + Efficiency + Sustainability + Environmental Protection

BPVC

- BPVC means **Boiler and Pressure Vessel Code**.
- It gives rules for the design, fabrication, inspection, and construction of boilers, pressure vessels, and nuclear components.
- It is used for equipment from small air tanks to power-plant and nuclear systems.

B31 Piping

- ASME B31 is the code for **pressure piping**.
- It covers piping used for water, steam, chemicals, fuel gas, hydrocarbons, refrigeration, and process systems.
- It ensures pipes can safely carry high-pressure or hazardous fluids.

A17 Elevators and Escalators

- ASME A17.1/CSA B44 is the **safety code for elevators and escalators**.
- It covers the design, installation, operation, inspection, testing, maintenance, alteration, and repair of these systems.
- This code helps prevent accidents such as drops, jams, and mechanical failures.

Easy exam line

- BPVC = **boilers and pressure vessels**.
- B31 = **piping systems**.
- A17 = **elevators and escalators**.
- All three are ASME standards that make engineering **safe, reliable, and standardized**.

ASCE Code of Ethics

1. Introduction

- ASCE stands for **American Society of Civil Engineers**.
- It provides a **Code of Ethics** for civil engineers.
- The modern code uses a **stakeholder-based model** and gives highest priority to **public safety, health, and welfare**.

2. Definition

- The ASCE Code of Ethics is a set of professional rules that guide civil engineers to act with **honesty, responsibility, competence, and public welfare** in mind.

3. Explanation of Concept

- The code says engineers must protect the public first.
- It also requires care for the **natural and built environment**, then the profession, clients, employers, and peers.
- It changed from old “canons” to a clearer **stakeholder hierarchy** in 2020.

4. Features / Types

- **Society first:** protect life, health, and safety.
- **Environment second:** support sustainable development.
- **Profession third:** maintain honor, dignity, and integrity.
- **Clients and employers fourth:** act as faithful agents and avoid conflicts.
- **Peers fifth:** respect colleagues and give credit fairly.

2. The Stakeholder Hierarchy (The Five Pillars)

The ASCE Code organizes the duties of an engineer into five categories. In the event of a conflict between duties, they are ranked in order of **priority**:

Priority	Stakeholder	Key Ethical Duty
1 (Highest)	Society	Protect health, safety, and welfare; ensure sustainable development.
2	Environment	Adhere to principles of sustainable development and protect the ecosystem.
3	Profession	Uphold the honor, integrity, and dignity of engineering; zero tolerance for bribery.
4	Clients/Employers	Act as faithful agents; maintain confidentiality; avoid conflicts of interest.
5	Peers	Treat colleagues with respect; give credit where it is due; mentor future engineers.

5. Importance

- It builds **public trust** in civil engineering.
- It prevents unsafe design, corruption, and unethical behavior.

- It supports **sustainable and ethical infrastructure development**.

6. Examples / Applications

- Stop a project if public safety is at risk.
- Report unsafe engineering practices.
- Refuse bribery or unfair contracts.
- Disclose conflicts of interest.
- Use honest technical reports and testimony.

7. Ethical Issues / Advantages

- Ethical issues include **conflict of interest, bribery, false reporting, and unsafe decisions**.
- Advantages include **public confidence, professional dignity, safer projects, and sustainable development**.

8. Conclusion

- ASCE Code of Ethics guides engineers to act as **trustworthy professionals**.
- Its main message is: **public safety first, environment next, and honesty always**.

Confidentiality

1. Introduction

- Confidentiality is an important ethical duty in professional work.
- It is especially important in engineering, computing, healthcare, and business.

2. Definition

- Confidentiality means **protecting sensitive information from unauthorized access, use, or disclosure**.

3. Explanation of Concept

- It means private, personal, technical, or business information must be kept secure.
- Only authorized people should access such information.
- It helps maintain trust between professionals, clients, and organizations.

4. Features / Types

- Personal data: name, address, Aadhaar, medical or financial details.
- Proprietary business data: trade secrets, plans, pricing, blueprints.
- Technical data: source code, security architecture, encryption keys.
- Legal data: contracts, reports, NDAs.

5. Importance

- Builds trust in professional relationships.

- Protects privacy and business advantage.
- Prevents misuse of data.
- Helps follow legal rules and avoid penalties.

6. Examples / Applications

- Protect client records.
- Keep design files and source code secure.
- Do not share passwords or internal emails.
- Do not discuss confidential project details in public places.

7. Ethical Issues / Advantages

- Ethical issues include data leaks, password sharing, forwarding private emails, and using company information for personal gain.
- Advantages include privacy protection, legal compliance, and stronger trust.

8. Conclusion

- Confidentiality is the duty to keep sensitive information safe.
- It is essential for ethical and responsible professional conduct.

Professional Integrity

1. Introduction

- Professional integrity is another key ethical value in engineering and all professions.
- It means behaving honestly and responsibly in all professional actions.

2. Definition

- Professional integrity means **acting with honesty, fairness, accountability, transparency, and moral consistency**.

3. Explanation of Concept

- A person with integrity does the right thing even under pressure.
- They give truthful reports, accept responsibility, and avoid cheating or corruption.
- Integrity is what gives a profession public trust and respect.

4. Features / Types

- Honesty.
- Transparency.
- Accountability.
- Reliability.
- Ethical behavior in decisions and actions.

5. Importance

- Improves credibility at the workplace.

- Builds trust among clients, colleagues, and supervisors.
- Prevents corruption and misconduct.
- Upholds the reputation of the profession.

6. Examples / Applications

- Reporting project progress truthfully.
- Admitting mistakes promptly.
- Not hiding safety defects.
- Not manipulating data to meet deadlines.
- Following professional codes like ACM or IEEE.

7. Ethical Issues / Advantages

- Ethical issues include false reporting, data manipulation, hiding flaws, and misuse of authority.
- Advantages include trust, quality work, fairness, and strong professional reputation.

8. Conclusion

- Professional integrity means always acting honestly and responsibly.
- It is essential for ethical engineering and public confidence.

If you want, I can now make both of these into a **very short 5-mark answer** or a **difference table: Confidentiality vs Integrity**.

give me as a example

Completed 2 steps

The correct phrase is **“give me an example”**, not “give me a example”.

Example

- **Give me an example of confidentiality.**
- **Example:** An engineer does not share client design files with outsiders.
- **Give me an example of integrity.**
- **Example:** A manager reports a safety defect honestly, even if it delays the project.

Easy memory tip

- Use **an** before example because **example** starts with a vowel sound.

Would you like **one example each for confidentiality and integrity** in very simple exam language?

Whistleblowing

1. Introduction

- Whistleblowing is an important ethical action that helps expose wrongdoing in organizations.

- It protects the public, supports honesty, and prevents serious harm.

2. Meaning

- Whistleblowing means **reporting unethical, illegal, or harmful activities** inside an organization to the proper authority.
- A whistleblower is a person who exposes corruption, fraud, safety violations, pollution, data breaches, harassment, or discrimination.

3. Need / Importance

- It protects public interest by preventing harm to society.
- It increases transparency and accountability in organizations.
- It can help prevent disasters caused by ignored warnings, such as engineering failures.
- It supports ethical culture and legal compliance.

4. Types

- **Internal whistleblowing:** reporting to supervisors, HR, or ethics committees.
- **External whistleblowing:** reporting to government, police, media, or regulators.
- **Anonymous whistleblowing:** identity is protected.
- **Personal whistleblowing:** about issues affecting the whistleblower directly.
- **Professional whistleblowing:** about harm to organization, public, or environment.

5. Risks Involved

- **Retaliation:** job loss, demotion, harassment, or negative reviews.
- **Social isolation:** colleagues may avoid the whistleblower.
- **Legal risks:** lawsuits or confidentiality disputes.
- **Emotional stress:** anxiety, fear, and mental pressure.
- **Career damage:** loss of promotion or future jobs.

6. Examples

- Reporting financial fraud or embezzlement.
- Exposing unsafe engineering practices.
- Reporting environmental pollution.
- Reporting data privacy violations.
- Reporting harassment, bribery, or corruption.

7. Ethical Significance

- Whistleblowing shows moral courage.
- It protects public welfare and discourages misuse of power.
- It is a duty for engineers, doctors, auditors, and technologists when public safety is at risk.

8. Protection for Whistleblowers

- India has the **Whistle Blowers Protection Act, 2014**.

- Other supporting rules include Companies Act provisions and SEBI/RBI guidelines.
- Global examples include the **Sarbanes-Oxley Act** and **Dodd-Frank Act**.

9. Conclusion

- Whistleblowing means reporting wrongdoing for the public good.
- It is necessary for safety, honesty, and accountability.
- Even though it has personal and professional risks, it is a vital ethical duty

Comparison of Codes of Ethics in Engineering

Aspect	ASME	ASCE	IEEE	IEI
Full form	American Society of Mechanical Engineers	American Society of Civil Engineers	Institute of Electrical and Electronics Engineers	Institution of Engineers (India)
Main focus	Mechanical systems, safety, competence, honesty	Public safety, sustainable development, environment	Public welfare, integrity, honesty, responsible conduct	Truth, honesty, justice, trustworthiness, human welfare, environment
Core principle	Use knowledge for human welfare and act honestly	Hold paramount public safety and welfare	Promote safety, honesty, and responsible professional behavior	Safeguard human life, welfare, and environment
Public safety	Very important	Highest priority	Very important	Very important
Environment	Implicit in responsible engineering	Strong emphasis on sustainable development and environment	Supports societal welfare and responsible technology use	Clearly includes environment and welfare
Integrity	Honesty, impartiality, no unfair competition	Integrity, honor, dignity, and zero tolerance for corruption	Integrity and ethical conduct are central	Truth, honesty, trustworthiness
Conflict of interest	Avoid conflicts and act faithfully	Avoid conflicts; act as faithful agents/trustees	Avoid conflicts and act responsibly	Act in public good and maintain fairness
Competence	Work only within competence; improve skills	Services only in areas of competence	Maintain competence and accept responsibility	Competence and accountability stressed
Corruption/bribery	No unfair competition or unethical practice	Zero tolerance for bribery, fraud, corruption	Ethical and responsible conduct; no misconduct	Justice, trustworthiness, anti-corruption values

Aspect	ASME	ASCE	IEEE	IEI
Professional growth	Increase competence and prestige	Continue learning and support profession	Encourage professional responsibility and growth	Uphold ethics and professional development

Main differences

- **ASME** focuses on mechanical engineering practice, safety, competence, and fair conduct.
- **ASCE** gives the strongest emphasis to **public safety and sustainable development**.
- **IEEE** emphasizes **integrity, responsible behavior, and public welfare** in electronics and computing.
- **IEI** highlights **truth, honesty, justice, trustworthiness, human welfare, and environmental care**.

Common points

- All four codes stress **public welfare, honesty, responsibility, and professional competence**.
- All reject unethical conduct and expect engineers to act in the interest of society.

Easy conclusion

- **ASME** = mechanical safety and competence.
- **ASCE** = public safety and sustainability.
- **IEEE** = integrity and responsible technology.
- **IEI** = honesty, trust, welfare, and environment.